License Plate Detection & Blurring using YOLOv8

Case Study

```
import cv2
print("OpenCV version:", cv2.__version__)

OpenCV version: 4.13.0
```

1. Business Problem & Context

With the rapid growth of camera-based traffic monitoring and surveillance systems, vast amounts of vehicle images and video streams are captured daily. These visual datasets frequently expose **Personally Identifiable Information (PII)**, particularly **vehicle license plate numbers**.

Uncontrolled exposure of such information leads to:

- Privacy violations
- Legal and regulatory risks (e.g., GDPR compliance)
- Ethical concerns in data usage

```
import os
import cv2
import numpy as np
import matplotlib.pyplot as plt
from ultralytics import YOLO

WARNING Ultralytics settings reset to default values. This may be due
to a possible problem with your settings or a recent ultralytics
package update.
View Ultralytics Settings with 'yolo settings' or at 'C:\Users\Rijul
Bhardwaj\AppData\Roaming\Ultralytics\settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings
runs_dir=path/to/dir'. For help see
https://docs.ultralytics.com/quickstart/#ultralytics-settings.
```

---

2. Dataset Overview

The dataset is structured into **training, validation, and test splits**, each containing images and corresponding YOLO-format annotations.

```
License Plate Detection and Blurring/
├── images/
│   ├── train/
│   ├── val/
│   └── test/
├── labels/
│   ├── train/
│   ├── val/
│   └── test/
```

Dataset Statistics

Split	Number of Images
Train	25,470
Validation	1,000
Test	400

Each label file follows the YOLO format:

```
<class_id> <center_x> <center_y> <width> <height>
```

All values are **normalized between 0 and 1**.

```
DATASET_PATH = r"C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at Scaler\Case Study License Plate Detection & Blurring using YOLOv8\License Plate Detection and Blurring"

splits = ["train", "val", "test"]

for split in splits:
    img_path = os.path.join(DATASET_PATH, "images", split)
    lbl_path = os.path.join(DATASET_PATH, "labels", split)

    print(f"\nChecking {split.upper()} split")

    if not os.path.exists(img_path):
        print(" Images folder missing")
        continue

    if not os.path.exists(lbl_path):
        print(" Labels folder missing")
        continue

    print(f" Images: {len(os.listdir(img_path))}")
    print(f" Labels: {len(os.listdir(lbl_path))}")
```

```
Checking TRAIN split
 Images: 23681
```

```
 Labels: 23379

Checking VAL split
 Images: 999
 Labels: 991

Checking TEST split
 Images: 374
 Labels: 373
```

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3. Converting YOLO Normalized Bounding Boxes to Pixel Coordinates

Why Conversion Is Required

YOLO predicts bounding boxes in **normalized coordinates**, while OpenCV operations (drawing, cropping, blurring) require **absolute pixel coordinates**. Therefore, coordinate transformation is mandatory.

Mathematical Transformation

Let:

- `img_w` = image width (pixels)
- `img_h` = image height (pixels)

Then:

- `center_x_px = center_x × img_w`
- `center_y_px = center_y × img_h`
- `width_px = width × img_w`
- `height_px = height × img_h`

Bounding box corners:

- `x_min = center_x_px - width_px / 2`
- `y_min = center_y_px - height_px / 2`
- `x_max = center_x_px + width_px / 2`
- `y_max = center_y_px + height_px / 2`

Code Implementation

```python
def yolo_to_pixel(label, img_w, img_h):
    """
    Converts YOLO normalized bounding box to pixel coordinates.
    label format: [class_id, center_x, center_y, width, height]
    """
```

```
    _, cx, cy, w, h = label

    cx *= img_w
    cy *= img_h
    w  *= img_w
    h  *= img_h

    x1 = int(cx - w / 2)
    y1 = int(cy - h / 2)
    x2 = int(cx + w / 2)
    y2 = int(cy + h / 2)

    return x1, y1, x2, y2
```

Business Impact

Accurate coordinate conversion is **critical for privacy compliance**. Any error at this stage can result in:

- Partial blurring (PII leakage)
- Excessive blurring (loss of contextual information)

```
def yolo_to_pixel(label, img_w, img_h):
    """
    YOLO label format:
    [class_id, center_x, center_y, width, height]
    """
    _, cx, cy, w, h = label

    cx *= img_w
    cy *= img_h
    w  *= img_w
    h  *= img_h

    x1 = int(cx - w / 2)
    y1 = int(cy - h / 2)
    x2 = int(cx + w / 2)
    y2 = int(cy + h / 2)

    return x1, y1, x2, y2
```

---

4. Overlaying Bounding Boxes on Raw Images

Why This Step Is Critical Before Training

Overlaying bounding boxes on raw images acts as a **pre-training validation step**. It allows us to:

- Verify annotation correctness

- Detect misaligned or missing labels
- Identify overly loose or overly tight bounding boxes

Training on incorrect annotations directly degrades detection quality and blurring accuracy.

Bounding Box Overlay Function

```
import cv2

def draw_bounding_box(image, bbox, color=(0, 255, 0), thickness=2):
    x1, y1, x2, y2 = bbox
    return cv2.rectangle(image, (x1, y1), (x2, y2), color, thickness)
```

Visualization Example

```
import matplotlib.pyplot as plt

img = cv2.imread("sample.jpg")
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

bbox = (x1, y1, x2, y2)
img_boxed = draw_bounding_box(img, bbox)

plt.imshow(img_boxed)
plt.axis("off")

def draw_bounding_box(image, bbox):
    x1, y1, x2, y2 = bbox
    return cv2.rectangle(image, (x1, y1), (x2, y2), (0, 255, 0), 2)

sample_img = os.listdir(os.path.join(DATASET_PATH, "images", "train"))
[0]
sample_lbl = sample_img.replace(".jpg", ".txt")

img = cv2.imread(os.path.join(DATASET_PATH, "images", "train",
sample_img))
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

with open(os.path.join(DATASET_PATH, "labels", "train", sample_lbl))
as f:
    label = list(map(float, f.readline().split()))

bbox = yolo_to_pixel(label, img.shape[1], img.shape[0])
img_boxed = draw_bounding_box(img, bbox)

plt.figure(figsize=(6,4))
plt.imshow(img_boxed)
plt.axis("off")

(np.float64(-0.5), np.float64(1023.5), np.float64(767.5), np.float64(-
0.5))
```

---

5. Impact of Annotation Quality on Blurring Accuracy

Annotation quality has a **direct impact** on the effectiveness of the privacy-preserving system.

Annotation Quality	Detection Result	Blurring Outcome
Tight & consistent boxes	High IoU	Accurate plate-only blur
Loose boxes	Lower IoU	Background over-blurred
Missing annotations	False negatives	Plate not blurred
Inconsistent labeling	Poor generalization	Unstable performance

> **Key Insight:** The model learns exactly what it is shown. High-quality annotations are essential for reliable anonymization.

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6. Image Preprocessing Strategy

License plates are **small, text-dense objects**, so preprocessing must preserve fine-grained details.

Preprocessing Decisions

- Resize images to **640 × 640** (YOLOv8 default)
- Preserve aspect ratio using letterboxing
- Avoid aggressive blurring or compression
- Allow YOLOv8 to handle normalization internally

Why This Works

- Maintains sharp edges and characters
- Improves small-object localization
- Prevents loss of critical visual information

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7. Why YOLOv8 Over a Two-Stage Detector

Comparison

Aspect	YOLOv8	Two-Stage Detectors
Architecture	Single-stage	Proposal + classifier
Inference speed	Very fast	Slow
Real-time capability	Yes	Limited
Deployment complexity	Low	High

Business Justification

Traffic surveillance systems require **real-time performance** and **scalability**. YOLOv8 processes images in a **single forward pass**, making it ideal for live feeds and edge deployment.

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8. YOLOv8 Variant Selection

Chosen Model: YOLOv8-S

Variant	Speed	Accuracy	Verdict
YOLOv8-n	Very fast	Lower	Misses small plates
YOLOv8-s	**Balanced**	**High**	▯ Selected
YOLOv8-m	Slower	Higher	Overkill
YOLOv8-l/x	Very slow	Marginal gain	Not cost-effective

YOLOv8-s provides the best balance between **small-object detection accuracy** and **real-time inference speed**.

```
model = YOLO("yolov8s.pt")
```

```
/ultralytics/assets/releases/download/v8.4.0/yolov8s.pt to
'yolov8s.pt': 100% ━━━━━━━━━━━━ 21.5MB 11.2MB/s 1.9s1.8s<0.1s
```

```
import os

print(os.path.exists(
    r"C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at Scaler\Case
Study License Plate Detection & Blurring using YOLOv8\License Plate
```

```
Detection and Blurring\images\train"
))
print(os.path.exists(
    r"C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at Scaler\Case
Study License Plate Detection & Blurring using YOLOv8\License Plate
Detection and Blurring\images\val"
))
```

```
True
True
```

```
from ultralytics import YOLO

model = YOLO("yolov8s.pt")
```

---

9. Influence of License Plate Characteristics on Model Choice

License plates are:

- Small relative to the image
- Rectangular and structured
- High-contrast but visually dense

These properties require:

- Strong spatial localization
- Multi-scale feature extraction
- Moderate model capacity

YOLOv8-s satisfies these requirements without excessive computational cost.

```
results = model.predict(
    source=os.path.join(DATASET_PATH, "images", "test"),
    conf=0.4,
    save=False
)
```

```
image 1/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\003a5aaf6d17c917.jpg:
448x640 4 cars, 131.3ms
image 2/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\008637722500f239.jpg:
480x640 3 cars, 95.6ms
```

```
image 3/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\017527da8bfeb97d.jpg:
448x640 1 person, 2 cars, 1 truck, 74.5ms
image 4/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\02a6ef3d9bd68e91.jpg:
480x640 2 cars, 79.3ms
image 5/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\03b7b71e1ffcb7a8.jpg:
448x640 1 car, 74.1ms
image 6/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\044417ca6134604f.jpg:
448x640 4 cars, 69.7ms
image 7/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\057569768fd6303e.jpg:
480x640 5 cars, 81.0ms
image 8/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\064a8def3049d040.jpg:
448x640 4 cars, 2 trucks, 77.0ms
image 9/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\0673b967f8c68eec.jpg:
480x640 1 car, 81.2ms
image 10/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\06b024413ad385a7.jpg:
512x640 1 truck, 104.1ms
image 11/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\0727983dd5f9e4e6.jpg:
480x640 1 truck, 76.4ms
image 12/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\0787b0fa95f545a5.jpg:
576x640 1 person, 8 cars, 1 truck, 106.9ms
image 13/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\0801961485534636.jpg:
480x640 3 cars, 79.6ms
image 14/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\081f5a6bc61b9c48.jpg:
480x640 5 cars, 76.5ms
image 15/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
```

```
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\08481c03daf6f35d.jpg:
448x640 1 car, 74.4ms
image 16/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\091c033b2a7df15b.jpg:
480x640 1 car, 1 truck, 77.0ms
image 17/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\09453a7c716a9ef3.jpg:
480x640 2 persons, 1 bicycle, 3 cars, 72.7ms
image 18/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\0c756c9366a8cb10.jpg:
480x640 5 cars, 78.6ms
image 19/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\0cacb08195a3e2d7.jpg:
448x640 1 car, 72.1ms
image 20/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\0d6ca8553971fefd.jpg:
448x640 8 cars, 1 truck, 80.7ms
image 21/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\0e50ea14c4fc1353.jpg:
448x640 3 persons, 2 cars, 77.0ms
image 22/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\0ee91c4938b6e7ee.jpg:
640x480 3 cars, 1 stop sign, 93.6ms
image 23/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\0f0596b1c511e071.jpg:
576x640 3 cars, 1 motorcycle, 102.3ms
image 24/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\0f4bfc46402a9f52.jpg:
640x640 1 keyboard, 118.3ms
image 25/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\0fbd1b85fc01d2ae.jpg:
384x640 6 cars, 106.1ms
image 26/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\11b155ab5b3331cf.jpg:
384x640 1 person, 1 car, 69.1ms
image 27/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
```

```
License Plate Detection and Blurring\images\test\11f80fda2c38011c.jpg:
448x640 1 car, 76.6ms
image 28/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\125622a2508e3b1d.jpg:
416x640 4 cars, 108.2ms
image 29/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\128a59bca025bed6.jpg:
480x640 1 car, 77.0ms
image 30/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\12b7ea40074d3b20.jpg:
480x640 1 car, 82.4ms
image 31/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\12d3b6665f7e1cf6.jpg:
640x480 1 person, 3 cars, 104.6ms
image 32/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\133921bd26fff729.jpg:
640x416 1 person, 3 cars, 143.2ms
image 33/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\140e8d10ff02e7e7.jpg:
448x640 2 persons, 1 car, 115.7ms
image 34/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\1545c73bdecb3e2f.jpg:
352x640 3 cars, 152.4ms
image 35/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\16789af24af158d8.jpg:
640x640 1 person, 2 cars, 163.1ms
image 36/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\16c15b29b30148c2.jpg:
576x640 2 cars, 129.7ms
image 37/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\17585ef2efb5ef9b.jpg:
640x480 7 cars, 107.4ms
image 38/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\18098dc9ee6aad80.jpg:
448x640 2 persons, 1 car, 1 traffic light, 98.4ms
image 39/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\182268e1f8c6525f.jpg:
```

```
448x640 5 cars, 96.7ms
image 40/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\185d5dfa193c4ced.jpg:
448x640 1 car, 98.7ms
image 41/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\185fd6653258d3ed.jpg:
480x640 1 car, 4 traffic lights, 110.7ms
image 42/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\19eba8ac64eed194.jpg:
480x640 8 cars, 112.7ms
image 43/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\19ff847d0cd1c5ec.jpg:
448x640 1 car, 96.8ms
image 44/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\1a3e9d87d733ff7f.jpg:
448x640 1 car, 95.1ms
image 45/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\1ca115083156d72.jpg:
448x640 1 person, 1 car, 110.2ms
image 46/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\1db3793d7c84faa1.jpg:
480x640 1 car, 106.8ms
image 47/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\1e6e75488d7878aa.jpg:
480x640 1 car, 1 truck, 106.5ms
image 48/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\1ead26febde18ce9.jpg:
480x640 1 car, 104.8ms
image 49/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\1eb2ebab9cd7adf6.jpg:
480x640 1 car, 2 potted plants, 95.7ms
image 50/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\1edfbe8c28a86caf.jpg:
384x640 1 car, 83.5ms
image 51/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\1f0e643b125f00ec.jpg:
448x640 1 person, 5 cars, 92.9ms
```

```
image 52/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\1f1d5bc31e444cc1.jpg:
448x640 1 car, 1 traffic light, 102.7ms
image 53/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\1f34454c4c073e1b.jpg:
608x640 3 cars, 170.0ms
image 54/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\228bd52bbe043677.jpg:
480x640 1 person, 5 cars, 1 truck, 1 traffic light, 104.3ms
image 55/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\23c2c79a9febbf3e.jpg:
384x640 2 cars, 1 truck, 90.6ms
image 56/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\246a0e4264a39433.jpg:
480x640 1 car, 112.7ms
image 57/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\25d5c2fbd8b99662.jpg:
384x640 1 car, 1 truck, 112.7ms
image 58/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\26183d88c4f8012f.jpg:
480x640 1 car, 111.2ms
image 59/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\26192da5ce2afa62.jpg:
448x640 3 persons, 2 bicycles, 1 car, 104.2ms
image 60/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2619ec27a314e69c.jpg:
480x640 3 cars, 113.5ms
image 61/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2753536137929411.jpg:
512x640 1 person, 2 cars, 123.3ms
image 62/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\27798906120ea394.jpg:
480x640 6 cars, 130.2ms
image 63/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\288a95c01e43cd14.jpg:
480x640 1 truck, 128.4ms
image 64/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
```

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Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\28bd6c6b1050e055.jpg:
448x640 3 cars, 1 bus, 1 cat, 140.3ms
image 65/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\28d1fb27a6e42ee7.jpg:
480x640 2 persons, 1 train, 1 truck, 129.9ms
image 66/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\28e26a4eb646c67a.jpg:
384x640 1 car, 133.4ms
image 67/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2956bcbedfc167b6.jpg:
480x640 2 cars, 113.4ms
image 68/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2adddd0b09bb0f17.jpg:
480x640 1 car, 120.4ms
image 69/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2b97f5bf137ee8d1.jpg:
448x640 12 cars, 1 truck, 140.9ms
image 70/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2bd26c63ebf598b7.jpg:
448x640 1 person, 2 cars, 123.5ms
image 71/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2c09fab0221652c5.jpg:
480x640 1 person, 1 car, 118.7ms
image 72/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2d40d6e377c82ec9.jpg:
640x480 5 persons, 1 car, 1 train, 122.4ms
image 73/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2d6640fb230770fd.jpg:
448x640 3 cars, 120.2ms
image 74/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2e8a83cfb1afe7ba.jpg:
480x640 1 person, 3 cars, 119.7ms
image 75/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2e95d7a799e23e11.jpg:
352x640 2 cars, 89.2ms
image 76/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
```

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License Plate Detection and Blurring\images\test\2f3887024d298547.jpg:
480x640 1 person, 1 car, 107.8ms
image 77/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2f90aaa72744452d.jpg:
480x640 5 cars, 108.5ms
image 78/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\2fe1f00b77a110a0.jpg:
448x640 1 car, 104.1ms
image 79/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\302d636c896c263f.jpg:
448x640 2 persons, 2 cars, 2 trucks, 103.7ms
image 80/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\30b6cfb60bf44533.jpg:
640x640 4 cars, 1 truck, 151.5ms
image 81/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\315aead7766727b8.jpg:
608x640 1 car, 172.3ms
image 82/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\321ffb38656f7311.jpg:
480x640 5 cars, 2 traffic lights, 126.6ms
image 83/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\3241f09a8964ddfb.jpg:
480x640 4 cars, 1 truck, 128.7ms
image 84/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\326c3286bcb5b2b0.jpg:
448x640 1 person, 2 cars, 117.8ms
image 85/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\334f31e809ec9a77.jpg:
480x640 2 cars, 117.1ms
image 86/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\33e0333ff031c4fb.jpg:
640x640 1 car, 171.6ms
image 87/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\343326e127297379.jpg:
448x640 5 persons, 3 cars, 94.8ms
image 88/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\35875c388efcddf0.jpg:
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480x640 4 cars, 107.6ms
image 89/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\36d7b8b3cca3b0f5.jpg:
448x640 1 person, 9 cars, 2 trucks, 90.3ms
image 90/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\37ff77d13a54aac2.jpg:
448x640 1 car, 97.0ms
image 91/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\3d98e3713783e82d.jpg:
448x640 6 cars, 105.3ms
image 92/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\3e156de979eb0828.jpg:
480x640 1 car, 109.4ms
image 93/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\3e2306f2cf4b2c67.jpg:
544x640 1 car, 138.4ms
image 94/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\3ecd5af7c4f11963.jpg:
480x640 1 bus, 1 truck, 102.4ms
image 95/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\3ed000628c6a0587.jpg:
448x640 1 car, 92.5ms
image 96/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\3feecc5809a29627.jpg:
640x640 1 car, 130.7ms
image 97/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\415d64bf8cfe82f2.jpg:
448x640 1 car, 75.9ms
image 98/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\428a5131c8cff7da.jpg:
640x640 1 car, 106.1ms
image 99/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\43339d9cbca470e4.jpg:
480x640 4 cars, 84.0ms
image 100/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\43b1b0028dd2db8d.jpg:
480x640 1 car, 79.8ms
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image 101/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\44afab29f5fa0abf.jpg:
480x640 3 cars, 77.8ms
image 102/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\453a77009b27e253.jpg:
448x640 1 train, 76.1ms
image 103/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\4572990fd64bb6be.jpg:
512x640 4 persons, 3 cars, 1 truck, 85.1ms
image 104/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\460753acbd6e6dad.jpg:
480x640 4 cars, 84.1ms
image 105/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\46a7991da6461831.jpg:
448x640 1 car, 1 truck, 75.6ms
image 106/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\4793138df3c05610.jpg:
352x640 1 car, 62.4ms
image 107/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\485e5f37dd13ffab.jpg:
384x640 2 cars, 1 bus, 67.4ms
image 108/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\488722909dd9c0ac.jpg:
480x640 1 car, 80.0ms
image 109/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\49595029959689b7.jpg:
480x640 1 car, 81.8ms
image 110/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\4ac941b9393d00f9.jpg:
480x640 2 cars, 2 trucks, 79.9ms
image 111/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\4b3aba5dfb7a0492.jpg:
480x640 1 car, 80.4ms
image 112/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\4c04b488ddc48225.jpg:
384x640 1 car, 63.8ms
image 113/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
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Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\4cb48c8bf41b70a4.jpg:
480x640 1 car, 83.9ms
image 114/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\4df1448703257ff0.jpg:
480x640 1 person, 2 cars, 1 truck, 80.0ms
image 115/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\4e2cb95b9c509b10.jpg:
384x640 5 cars, 72.2ms
image 116/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\4e593c88022ff6b1.jpg:
480x640 1 person, 5 cars, 98.0ms
image 117/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\4ee3f0e9a2fb20a9.jpg:
448x640 1 person, 1 car, 2 motorcycles, 72.5ms
image 118/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\4fa24c1abe969cb7.jpg:
448x640 1 car, 1 motorcycle, 84.9ms
image 119/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\4ff8bebd6d6a0361.jpg:
480x640 1 bicycle, 1 car, 115.5ms
image 120/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\5021041c726b6ee4.jpg:
480x640 3 persons, 1 bus, 1 truck, 118.8ms
image 121/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\50631041f74001aa.jpg:
448x640 1 car, 130.1ms
image 122/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\508f0a75156e5fd4.jpg:
448x640 4 persons, 2 cars, 127.2ms
image 123/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\50c37aeaf19acd5b.jpg:
448x640 1 person, 1 bicycle, 5 cars, 1 motorcycle, 101.7ms
image 124/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\5290f4027491d09b.jpg:
480x640 1 person, 1 car, 117.2ms
image 125/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\52bf6b555e578a34.jpg:
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448x640 3 persons, 3 cars, 1 train, 99.4ms
image 126/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\52ceb1fc30b413e5.jpg:
480x640 6 persons, 2 cars, 1 truck, 100.2ms
image 127/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\53386dae3cf13cc3.jpg:
480x640 2 cars, 102.0ms
image 128/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\5373ec295ea62c47.jpg:
448x640 3 cars, 93.7ms
image 129/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\53925df03b471f5d.jpg:
480x640 2 cars, 107.4ms
image 130/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\53ad98b12752ad16.jpg:
480x640 8 cars, 129.6ms
image 131/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\548a8e997a66a0c4.jpg:
576x640 3 cars, 122.6ms
image 132/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\548a985825bb991f.jpg:
480x640 (no detections), 108.1ms
image 133/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\54bd93e5a4de808a.jpg:
480x640 1 car, 126.6ms
image 134/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\54ebca2064066a49.jpg:
448x640 1 car, 103.1ms
image 135/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\5581a1b7d2f0f2b4.jpg:
640x640 3 cars, 138.2ms
image 136/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\5641e6ad7d2600f2.jpg:
480x640 2 persons, 2 cars, 109.1ms
image 137/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\575d96d014e4fbab.jpg:
480x640 1 truck, 106.0ms
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image 138/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\593e594137f374ab.jpg:
448x640 1 car, 99.9ms
image 139/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\59b83c5f8f1dcfa4.jpg:
640x480 2 cars, 101.1ms
image 140/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\5a122adbb1a776b7.jpg:
480x640 9 cars, 102.9ms
image 141/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\5a96f5e1f9be8d42.jpg:
480x640 5 persons, 1 car, 92.7ms
image 142/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\5cc19b450a51ee4c.jpg:
480x640 2 cars, 103.1ms
image 143/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\5daafbcf76fa6602.jpg:
480x640 3 cars, 102.7ms
image 144/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\5df8816356fc2b29.jpg:
320x640 1 car, 90.0ms
image 145/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\5fb888d79b551331.jpg:
640x640 1 car, 164.0ms
image 146/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\5ff91ccfc6f15d04.jpg:
448x640 4 cars, 104.1ms
image 147/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\61a69ad713142e45.jpg:
480x640 1 person, 1 truck, 95.6ms
image 148/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\620bd952592bc0e9.jpg:
480x640 2 persons, 6 cars, 1 truck, 103.5ms
image 149/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\621837d55c229864.jpg:
480x640 1 person, 1 car, 1 parking meter, 1 bench, 102.2ms
image 150/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
```

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Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\628a23a43f19c4db.jpg:
384x640 1 person, 5 cars, 1 traffic light, 98.3ms
image 151/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\62bb93bbd270dd9a.jpg:
448x640 2 persons, 1 car, 172.2ms
image 152/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\6306fc7f3573eb70.jpg:
448x640 6 cars, 115.2ms
image 153/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\63d3df798bc8840f.jpg:
640x640 2 cars, 143.8ms
image 154/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\643c3cc1a4e85db2.jpg:
480x640 1 car, 106.2ms
image 155/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\647ff22cf6950211.jpg:
544x640 1 car, 112.4ms
image 156/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\6504632e0dc25997.jpg:
480x640 3 persons, 8 cars, 1 motorcycle, 2 buss, 1 truck, 1 traffic
light, 135.2ms
image 157/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\659dbf4f8c0fd29d.jpg:
480x640 3 cars, 109.1ms
image 158/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\665cac5fd1cc2186.jpg:
448x640 2 persons, 2 cars, 98.9ms
image 159/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\66771a4870c324f9.jpg:
384x640 1 person, 1 bicycle, 1 car, 1 cell phone, 86.4ms
image 160/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\66be6b583048fa94.jpg:
448x640 (no detections), 107.9ms
image 161/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\67945cf7a6beccdf.jpg:
384x640 5 cars, 97.8ms
image 162/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
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Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\67c834b73882a9f9.jpg:
448x640 5 persons, 3 cars, 100.4ms
image 163/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\6846c275ded01f85.jpg:
640x640 1 car, 1 truck, 2 traffic lights, 131.2ms
image 164/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\693acf98575cbe27.jpg:
480x640 1 car, 1 truck, 1 parking meter, 103.7ms
image 165/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\69450fa183d57a7b.jpg:
384x640 4 cars, 1 truck, 91.7ms
image 166/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\6a0ef049e5ec4b16.jpg:
448x640 (no detections), 100.6ms
image 167/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\6aff04e9c32aec0f.jpg:
640x480 1 car, 1 parking meter, 110.4ms
image 168/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\6b113b3edbadfe5d.jpg:
480x640 5 cars, 115.4ms
image 169/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\6c144e47161867c2.jpg:
384x640 1 car, 86.4ms
image 170/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\6e819dc674a74078.jpg:
480x640 3 cars, 102.9ms
image 171/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\6eb18d1ad17cd174.jpg:
448x640 1 person, 1 car, 2 trucks, 94.4ms
image 172/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\70e7b5173fe1e289.jpg:
480x640 1 car, 99.7ms
image 173/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\71881a9c0f7f0a86.jpg:
448x640 3 persons, 7 cars, 92.6ms
image 174/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
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License Plate Detection and Blurring\images\test\71dbb47ebe504abe.jpg:
352x640 1 car, 2 trucks, 75.6ms
image 175/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\724307be418b2ed2.jpg:
480x640 6 cars, 99.3ms
image 176/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\7275e72e1996267f.jpg:
512x640 3 cars, 105.3ms
image 177/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\73467682c5995b65.jpg:
512x640 3 persons, 6 cars, 1 truck, 108.1ms
image 178/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\747b0f12f54703ee.jpg:
480x640 2 cars, 101.8ms
image 179/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\749b4451492fc250.jpg:
448x640 2 cars, 1 truck, 99.0ms
image 180/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\74ee6d1b58ae2e70.jpg:
448x640 1 car, 86.2ms
image 181/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\7539902687e8f3a4.jpg:
480x640 4 cars, 2 trucks, 87.1ms
image 182/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\76e14d05ee8acb6d.jpg:
448x640 1 car, 81.7ms
image 183/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\77e716224c85410f.jpg:
480x640 3 cars, 1 stop sign, 81.6ms
image 184/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\791d63b1c3499e4f.jpg:
512x640 2 cars, 1 truck, 97.0ms
image 185/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\7a1f9520c7ecd5b1.jpg:
544x640 2 cars, 119.1ms
image 186/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\7b4e86e1c94d65de.jpg:
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480x640 1 truck, 90.2ms
image 187/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\7bcf6b4ec09f1ea7.jpg:
480x640 1 car, 82.0ms
image 188/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\7cbc7ee444622a66.jpg:
640x640 3 cars, 1 suitcase, 112.4ms
image 189/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\7f56adf4b9306ac9.jpg:
448x640 2 cars, 1 truck, 81.9ms
image 190/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\7f9740e95a74f2b9.jpg:
480x640 1 car, 86.5ms
image 191/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\7fc0fb6035bc06a6.jpg:
448x640 2 persons, 5 cars, 85.1ms
image 192/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\80d21c58a71a0751.jpg:
384x640 4 cars, 71.9ms
image 193/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\80e02a518ffe4cb2.jpg:
480x640 1 car, 87.7ms
image 194/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\8256e277c7f47797.jpg:
480x640 8 cars, 1 traffic light, 81.0ms
image 195/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\82b53fe7f9147c96.jpg:
448x640 1 car, 73.3ms
image 196/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\844adfe06e003c09.jpg:
480x640 1 car, 86.5ms
image 197/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\85b1dafd26aa98df.jpg:
448x640 2 persons, 12 cars, 75.5ms
image 198/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\85ddb7e0f60156a3.jpg:
448x640 2 persons, 7 cars, 1 truck, 81.8ms
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image 199/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\85f0b4db3eb512ce.jpg:
448x640 1 car, 75.2ms
image 200/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\86c92b0402fec141.jpg:
448x640 2 cars, 81.3ms
image 201/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\87793700161a7a6b.jpg:
448x640 1 person, 5 cars, 74.7ms
image 202/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\87b4f1202cd06440.jpg:
480x640 1 person, 4 cars, 106.9ms
image 203/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\88df55ff4e13b90d.jpg:
480x640 1 car, 80.0ms
image 204/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\88fc866f92860d70.jpg:
640x640 2 persons, 1 car, 1 truck, 125.4ms
image 205/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\894cacdb385fdb6b.jpg:
480x640 2 cars, 160.6ms
image 206/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\895d440e05b2a8d8.jpg:
480x640 1 car, 119.1ms
image 207/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\8c63cf76166c3bd7.jpg:
512x640 1 car, 120.8ms
image 208/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\8cb5553e75d3d311.jpg:
480x640 1 person, 4 cars, 116.3ms
image 209/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\8cca720c410ee7f8.jpg:
640x480 5 persons, 8 cars, 1 truck, 106.8ms
image 210/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\8cdcb833ef1ef049.jpg:
480x640 4 cars, 96.5ms
image 211/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
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Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\8ceed1ec9bc3ac45.jpg:
640x480 12 cars, 109.4ms
image 212/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\9034927e7438bdd6.jpg:
640x480 1 person, 4 cars, 120.9ms
image 213/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\90596bf3313e72e3.jpg:
352x640 3 cars, 86.6ms
image 214/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\908a3ac555a1a509.jpg:
448x640 1 car, 1 truck, 113.1ms
image 215/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\90abd4265adf5a00.jpg:
448x640 2 persons, 24 cars, 1 truck, 4 traffic lights, 125.2ms
image 216/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\911e694dcc0814bf.jpg:
480x640 9 cars, 102.4ms
image 217/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\9238f5062ff1dc8d.jpg:
480x640 1 car, 95.0ms
image 218/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\9286a99f243b359a.jpg:
512x640 1 car, 101.8ms
image 219/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\932eef14e5ee78b2.jpg:
448x640 1 truck, 95.7ms
image 220/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\934645379fe657e1.jpg:
480x640 1 car, 97.2ms
image 221/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\944281d0485869f0.jpg:
448x640 1 car, 90.0ms
image 222/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\9498d0aff8f9ff8c.jpg:
384x640 1 car, 77.8ms
image 223/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
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License Plate Detection and Blurring\images\test\955a8e4c8ba8116e.jpg:
480x640 2 cars, 1 truck, 95.6ms
image 224/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\98bcfbc4d3c8abbc.jpg:
480x640 3 cars, 99.6ms
image 225/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\99732f172ac80128.jpg:
384x640 2 cars, 1 truck, 79.7ms
image 226/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\9995dfc6e0eae5a1.jpg:
448x640 1 car, 85.7ms
image 227/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\99c9c657582f958f.jpg:
448x640 2 cars, 1 truck, 86.4ms
image 228/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\9a4fe6361597f264.jpg:
448x640 1 car, 87.7ms
image 229/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\9a795bbf4e94d630.jpg:
448x640 1 bus, 2 trucks, 85.9ms
image 230/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\9c92cadb0f6237c0.jpg:
448x640 1 person, 5 cars, 84.9ms
image 231/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\9ebbaf4d1f8d6e7b.jpg:
352x640 1 car, 73.0ms
image 232/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a21aa7c1d7b93d12.jpg:
480x640 1 car, 1 truck, 103.5ms
image 233/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a3ad91fabd188be3.jpg:
448x640 1 person, 4 cars, 96.9ms
image 234/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a52079c6cc4050cd.jpg:
480x640 1 car, 97.4ms
image 235/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a54e7bb80448f3b0.jpg:
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480x640 2 cars, 96.9ms
image 236/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a56e82026bb71ad5.jpg:
448x640 4 cars, 92.0ms
image 237/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a5d6f8d36672fa64.jpg:
448x640 1 bus, 96.3ms
image 238/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a67bc3f5f8650c73.jpg:
640x640 1 car, 122.7ms
image 239/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a6e3b0b73220cd32.jpg:
448x640 1 person, 1 car, 89.8ms
image 240/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a72a8cda1bb31e34.jpg:
480x640 1 car, 93.8ms
image 241/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a774a6f81fea258b.jpg:
480x640 1 car, 98.3ms
image 242/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a79296fb4a48245e.jpg:
448x640 3 persons, 3 cars, 90.4ms
image 243/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a823532e163f5722.jpg:
480x640 1 car, 1 bus, 100.2ms
image 244/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a929dc75c20da7d8.jpg:
640x640 1 car, 139.2ms
image 245/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\a9b2a2018f845393.jpg:
480x640 1 car, 1 potted plant, 124.4ms
image 246/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\aac1aaaf824b63dd.jpg:
480x640 1 car, 1 truck, 111.9ms
image 247/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\abca9401dcf8f7ba.jpg:
480x640 3 cars, 105.6ms
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image 248/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\aca8821ff0368720.jpg:
480x640 3 cars, 104.0ms
image 249/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\ad10c7d29f111692.jpg:
480x640 1 car, 94.5ms
image 250/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\ae6724423f0d111a.jpg:
512x640 10 persons, 1 car, 106.2ms
image 251/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\aefe747315dd79fc.jpg:
480x640 6 cars, 146.3ms
image 252/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\affd16c51644a893.jpg:
480x640 2 persons, 1 car, 120.5ms
image 253/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b01d46f9911d558b.jpg:
288x640 1 car, 96.9ms
image 254/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b076ad266891d7aa.jpg:
448x640 2 cars, 142.0ms
image 255/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b09813a8742277c7.jpg:
512x640 3 cars, 132.6ms
image 256/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b101900b26128253.jpg:
480x640 2 persons, 1 car, 1 truck, 110.3ms
image 257/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b1096bc91a89b0cf.jpg:
448x640 2 cars, 105.1ms
image 258/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b15d6c0bdf90226d.jpg:
480x640 1 car, 118.7ms
image 259/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b193070a9c45b5ab.jpg:
480x640 3 cars, 268.5ms
image 260/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
```

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License Plate Detection and Blurring\images\test\b1a50a3824887ee2.jpg:
448x640 2 persons, 3 cars, 2 trucks, 169.3ms
image 261/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b25d7d2cb7abf86b.jpg:
448x640 4 cars, 1 truck, 105.6ms
image 262/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b32671b7cc279583.jpg:
640x640 1 person, 1 car, 1 motorcycle, 112.5ms
image 263/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b3b61da98e22cd4a.jpg:
480x640 1 person, 2 cars, 85.0ms
image 264/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b57fe3aa2ff9577e.jpg:
384x640 1 car, 1 truck, 69.5ms
image 265/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b5e7183b6a5abe6c.jpg:
480x640 1 car, 1 truck, 1 dog, 83.5ms
image 266/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b6580dec5ada277d.jpg:
448x640 1 car, 80.8ms
image 267/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b6e55f97085c3732.jpg:
480x640 3 cars, 1 truck, 87.3ms
image 268/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b6ecda23586a6ba5.jpg:
480x640 2 cars, 90.5ms
image 269/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b7cdb5c0c99002b7.jpg:
448x640 1 person, 2 cars, 1 motorcycle, 80.7ms
image 270/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b8a3f2ea385e45b3.jpg:
384x640 4 persons, 7 cars, 8 traffic lights, 73.7ms
image 271/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b91c3aaba25bf914.jpg:
448x640 1 car, 79.8ms
image 272/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b94993a2d67b455e.jpg:
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448x640 2 cars, 87.7ms
image 273/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\b9f5b9acf1777acf.jpg:
480x640 3 cars, 86.8ms
image 274/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\bacc80fafc0566c3.jpg:
480x640 1 bus, 1 truck, 85.6ms
image 275/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\bbcac63e32bd8137.jpg:
448x640 1 car, 80.4ms
image 276/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\bd12ea92bcab1afd.jpg:
480x640 6 cars, 84.8ms
image 277/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\be654a7eabe0e891.jpg:
480x640 1 car, 79.0ms
image 278/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\be9fd0014b5a4f2a.jpg:
448x640 1 car, 73.5ms
image 279/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\bf0aac0878b0a3d2.jpg:
512x640 1 truck, 91.5ms
image 280/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\bf2f25f3ed9ff4d5.jpg:
448x640 1 car, 78.3ms
image 281/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\c1d8b110186e095a.jpg:
480x640 7 cars, 1 truck, 82.7ms
image 282/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\c274e026ac907eb9.jpg:
480x640 1 car, 80.5ms
image 283/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\c50d184afad1a9dc.jpg:
480x640 1 bus, 83.4ms
image 284/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\c650ff8d3e8e75b3.jpg:
416x640 4 cars, 84.5ms
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image 285/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\c7b1e6f7c38fa1a0.jpg:
480x640 1 person, 8 cars, 1 truck, 3 traffic lights, 79.0ms
image 286/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\c7df5232291486a3.jpg:
640x640 1 truck, 107.2ms
image 287/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\c91ee912164d8ecb.jpg:
480x640 2 cars, 76.5ms
image 288/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\cb4b754537798d23.jpg:
448x640 3 cars, 1 truck, 74.2ms
image 289/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\cb8c75fc1c7ccf73.jpg:
640x640 3 cars, 3 traffic lights, 108.0ms
image 290/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\ccc1a2d44a290368.jpg:
448x640 1 car, 1 truck, 75.0ms
image 291/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\cdb1e8621d1c624e.jpg:
448x640 1 car, 72.1ms
image 292/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\ce97f7bc90e97109.jpg:
448x640 5 cars, 1 truck, 75.9ms
image 293/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d027c6e32db60e3c.jpg:
480x640 2 cars, 79.4ms
image 294/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d27e094e98374ff8.jpg:
448x640 3 cars, 77.6ms
image 295/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d28d71c2690c16ed.jpg:
480x640 12 cars, 86.0ms
image 296/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d2fe2b47668e9d8e.jpg:
480x640 2 cars, 1 motorcycle, 80.7ms
image 297/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
```

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Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d3729058c6a4186e.jpg:
480x640 3 persons, 2 cars, 79.3ms
image 298/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d44c0b2252bf8a92.jpg:
480x640 1 car, 82.2ms
image 299/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d5058059dc0ce0ad.jpg:
480x640 2 cars, 1 bus, 1 dog, 83.2ms
image 300/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d53a3bc813477685.jpg:
384x640 1 car, 65.9ms
image 301/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d565d93637d4e76d.jpg:
448x640 1 car, 82.7ms
image 302/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d59b7a0ff294e3be.jpg:
576x640 1 car, 1 truck, 96.2ms
image 303/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d5f4069e6734ac06.jpg:
480x640 3 cars, 80.9ms
image 304/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d648b2ae1abc09f3.jpg:
544x640 1 person, 4 cars, 87.9ms
image 305/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d6c5271e96ec1a61.jpg:
640x480 3 cars, 1 parking meter, 84.3ms
image 306/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d7b71c9fd144d58d.jpg:
640x640 1 car, 108.7ms
image 307/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d7d49b1a3706f3eb.jpg:
448x640 2 cars, 72.8ms
image 308/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d830c3573e57bfc0.jpg:
480x640 2 persons, 5 cars, 80.7ms
image 309/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
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License Plate Detection and Blurring\images\test\d8cc6d566d01054b.jpg:
640x640 (no detections), 100.9ms
image 310/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d8daed582e6cce2d.jpg:
480x640 1 car, 81.4ms
image 311/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d8f6c135ec5486ff.jpg:
448x640 1 car, 75.5ms
image 312/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\d9fa2abf3719a4bd.jpg:
480x640 2 cars, 84.2ms
image 313/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\da1352ab6054a2e4.jpg:
480x640 (no detections), 108.9ms
image 314/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\da8186b39f042cdf.jpg:
480x640 3 cars, 142.0ms
image 315/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\daac3a6c64a79bea.jpg:
640x512 2 persons, 7 cars, 1 truck, 146.5ms
image 316/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\de2b5afb7eda96cf.jpg:
448x640 4 cars, 108.0ms
image 317/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\dec4608715b2d28d.jpg:
384x640 2 cars, 92.0ms
image 318/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\defd8f4b30b3e1e1.jpg:
480x640 6 persons, 2 cars, 108.6ms
image 319/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e08424f3cd5f6dac.jpg:
480x640 1 car, 129.7ms
image 320/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e1120bfe48f85c1b.jpg:
576x640 5 cars, 219.0ms
image 321/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e1925cd0ae28fbd9.jpg:
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416x640 5 cars, 108.2ms
image 322/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e2427db217e5c3db.jpg:
448x640 2 cars, 111.2ms
image 323/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e40cafe0f0d3a550.jpg:
480x640 1 car, 110.1ms
image 324/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e4d600a63db8bf3a.jpg:
640x640 1 bus, 1 truck, 130.9ms
image 325/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e52d4f347b2cf90f.jpg:
480x640 5 cars, 107.3ms
image 326/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e57d38ae6a921518.jpg:
480x640 1 person, 6 cars, 130.2ms
image 327/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e5edf93b7c9f5edb.jpg:
480x640 1 person, 1 car, 107.0ms
image 328/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e6299b9e04680adb.jpg:
480x640 2 persons, 8 cars, 99.3ms
image 329/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e742e284c7c9da96.jpg:
480x640 6 cars, 95.8ms
image 330/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e79f6777f2fc08b2.jpg:
448x640 4 cars, 93.3ms
image 331/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e82f13b4a2fe69f3.jpg:
480x640 1 car, 99.9ms
image 332/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e8b36c888a75d742.jpg:
480x640 3 persons, 11 cars, 103.7ms
image 333/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e9561d7c16db565f.jpg:
448x640 3 cars, 1 bus, 113.2ms
```

```
image 334/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e9cd346a4a84d594.jpg:
512x640 1 car, 1 bus, 144.2ms
image 335/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\e9fdd0a41cb61da6.jpg:
480x640 2 cars, 1 truck, 1 traffic light, 146.3ms
image 336/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\ea02d99ba51372ee.jpg:
480x640 10 cars, 123.8ms
image 337/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\ebaefbcb87bfa8a1.jpg:
448x640 10 persons, 1 car, 109.5ms
image 338/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\ec8a36f7874c0c34.jpg:
480x640 4 cars, 105.2ms
image 339/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\ed7d99098d4af8e6.jpg:
480x640 1 car, 109.2ms
image 340/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\edecf3a6d569b7c9.jpg:
640x448 1 car, 115.9ms
image 341/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\ee047c8ca2cca8a2.jpg:
480x640 4 persons, 2 cars, 1 truck, 108.9ms
image 342/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\ef1bd7b718a3f6e8.jpg:
480x640 2 persons, 1 car, 1 motorcycle, 106.5ms
image 343/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\efde2a25c1c9c924.jpg:
448x640 1 car, 96.2ms
image 344/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f0c7057612710f21.jpg:
384x640 1 car, 80.2ms
image 345/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f1131b93a33cefb9.jpg:
448x640 2 cars, 94.3ms
image 346/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
```

```
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f194eaf4f3d1d835.jpg:
480x640 1 person, 3 cars, 102.6ms
image 347/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f3814f4a6121838d.jpg:
320x640 4 cars, 75.9ms
image 348/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f38265abb22a00e4.jpg:
448x640 9 cars, 1 truck, 93.0ms
image 349/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f3bd3adea149193b.jpg:
448x640 2 cars, 1 motorcycle, 103.2ms
image 350/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f3c9af100984744f.jpg:
480x640 1 person, 10 cars, 109.7ms
image 351/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f3d472955c13cae0.jpg:
512x640 7 cars, 96.6ms
image 352/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f5128e7a123b4fa8.jpg:
448x640 1 car, 1 truck, 80.8ms
image 353/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f53243c2bb551b8a.jpg:
448x640 1 car, 91.4ms
image 354/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f5c2a80a21ddb78c.jpg:
448x640 1 car, 84.2ms
image 355/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f5d1729aa333b284.jpg:
480x640 2 cars, 94.2ms
image 356/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f6256bceaf66693f.jpg:
480x640 1 car, 99.0ms
image 357/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f700c5b69c973db6.jpg:
448x640 1 car, 90.5ms
image 358/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
```

```
License Plate Detection and Blurring\images\test\f73b754cdb1ab677.jpg:
448x640 1 car, 75.5ms
image 359/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f83d18919353c483.jpg:
448x640 11 persons, 2 cars, 1 truck, 81.6ms
image 360/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f9170e8c13a99991.jpg:
544x640 1 truck, 1 dog, 92.3ms
image 361/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\f9f539977bfea25e.jpg:
448x640 2 persons, 5 cars, 80.0ms
image 362/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\fa897478280a2758.jpg:
480x640 1 person, 4 cars, 1 traffic light, 90.0ms
image 363/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\fa9147596edc058f.jpg:
448x640 1 car, 1 truck, 78.4ms
image 364/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\fb4e60b5cee8b88e.jpg:
640x416 5 cars, 4 traffic lights, 84.5ms
image 365/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\fb55b73f241bf50a.jpg:
448x640 1 car, 79.1ms
image 366/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\fc0a607ba5f38505.jpg:
448x640 1 car, 80.4ms
image 367/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\fc0f46431cb1dbe9.jpg:
416x640 5 persons, 3 cars, 78.6ms
image 368/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\fd07d2db70cf53d5.jpg:
480x640 1 car, 84.2ms
image 369/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\fdd963edc28bf163.jpg:
480x640 2 cars, 78.7ms
image 370/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\fe6139d150a3e2a8.jpg:
```

```
640x640 4 cars, 4 boats, 106.9ms
image 371/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\feaf383fc5de383f.jpg:
480x640 2 cars, 85.6ms
image 372/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\fed5c0275ff3a440.jpg:
448x640 2 cars, 1 truck, 73.7ms
image 373/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\ff69d090c735eccb.jpg:
480x640 2 cars, 83.9ms
image 374/374 C:\Users\Rijul Bhardwaj\OneDrive\Desktop\Projects at
Scaler\Case Study License Plate Detection & Blurring using YOLOv8\
License Plate Detection and Blurring\images\test\ff85b09876d61631.jpg:
640x544 1 person, 3 cars, 107.3ms
Speed: 2.9ms preprocess, 100.6ms inference, 0.8ms postprocess per
image at shape (1, 3, 640, 544)
```

```python
def blur_license_plate(image, bbox):
    x1, y1, x2, y2 = bbox
    roi = image[y1:y2, x1:x2]
    blurred_roi = cv2.GaussianBlur(roi, (51, 51), 0)
    image[y1:y2, x1:x2] = blurred_roi
    return image
```

---

10. Applying Blurring Only Within the Detected Bounding Box

To preserve non-sensitive visual information, blurring is applied **only inside the detected license plate region**.

Blurring Function

```python
def blur_license_plate(image, bbox):
    x1, y1, x2, y2 = bbox
    roi = image[y1:y2, x1:x2]
    blurred_roi = cv2.GaussianBlur(roi, (51, 51), 0)
    image[y1:y2, x1:x2] = blurred_roi
    return image
```

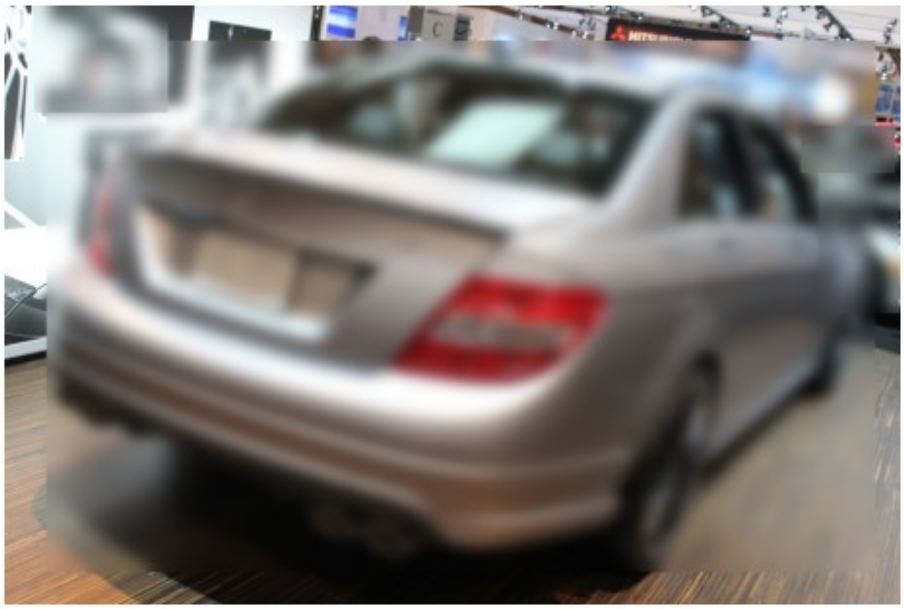
Why This Matters

- Ensures complete anonymization of PII
- Preserves surrounding scene context
- Maintains analytical usefulness of the image

```
for r in results:
    img = r.orig_img.copy()

    for box in r.boxes.xyxy.cpu().numpy():
        x1, y1, x2, y2 = map(int, box)
        img = blur_license_plate(img, (x1, y1, x2, y2))

    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    plt.figure(figsize=(6,4))
    plt.imshow(img)
    plt.axis("off")
    break
```



---

11. Final Insights & Recommendations

- **Annotation quality** is the most critical factor for effective anonymization
- **YOLOv8** enables scalable, real-time privacy protection
- Pretrained models are viable when hardware constraints exist
- This system should be deployed as a **mandatory preprocessing step** before storing or sharing visual data

---

12. Conclusion

This project demonstrates an **end-to-end, privacy-preserving computer vision pipeline** for license plate anonymization. By combining high-quality annotations, accurate coordinate transformations, efficient detection, and region-based blurring, the system achieves both **regulatory compliance** and **operational efficiency**.

Business Questions

1. How do you convert normalized YOLO bounding box coordinates into pixel coordinates for OpenCV?

Explanation

YOLO outputs bounding boxes in **normalized coordinates**, where values lie in the range ([0,1]) relative to image dimensions:

```
(center_x, center_y, width, height)
```

However, OpenCV operates in **absolute pixel coordinates**. Therefore, normalized values must be scaled using the image width and height.

Conversion Logic

Let:

- `img_w` = image width in pixels
- `img_h` = image height in pixels

Steps:

- Convert center coordinates to pixels
- Convert width and height to pixels
- Compute top-left and bottom-right corners

Code

```python
def yolo_to_pixel(label, img_w, img_h):
    """
    Converts YOLO normalized coordinates to pixel coordinates.
    label format: [class_id, center_x, center_y, width, height]
    """
    _, cx, cy, w, h = label

    cx *= img_w
    cy *= img_h
    w  *= img_w
    h  *= img_h

    x1 = int(cx - w / 2)
    y1 = int(cy - h / 2)
    x2 = int(cx + w / 2)
    y2 = int(cy + h / 2)
```

```
    return x1, y1, x2, y2
```

Business Importance

Accurate coordinate conversion is critical because any error directly affects:

- Bounding box alignment
- Blurring precision
- Privacy compliance

---

2. Write a function to overlay bounding boxes on raw images. Why is this step critical before training?

Bounding Box Overlay Function

```
import cv2

def draw_bounding_box(image, bbox, color=(0, 255, 0), thickness=2):
    x1, y1, x2, y2 = bbox
    return cv2.rectangle(image, (x1, y1), (x2, y2), color, thickness)
```

Why This Step Is Critical

Overlaying bounding boxes on raw images is a **pre-training validation step** that ensures:

- Annotations correctly align with license plates
- No missing or shifted labels exist
- Bounding boxes are neither too loose nor too tight

Business Impact

Training on incorrect annotations leads to:

- Poor detection accuracy
- Incomplete or excessive blurring
- Potential exposure of sensitive information

This step significantly reduces downstream privacy risks.

---

3. How did annotation quality impact the final blurring accuracy?

Annotation quality had a **direct and measurable impact** on blurring performance.

Observed Impact

Annotation Quality	Detection Result	Blurring Outcome
Tight, consistent boxes	High IoU	Accurate plate-only blur
Loose boxes	Lower IoU	Background blurred
Missing labels	False negatives	Plate not blurred
Inconsistent style	Poor generalization	Unstable blur

Key Insight

> The model learns exactly what it is shown. High-quality annotations are essential for reliable anonymization.

---

4. How did you preprocess images before feeding them into YOLOv8 without losing fine-grained plate details?

License plates are **small, text-dense objects**, so preprocessing must be conservative.

Preprocessing Strategy

- Resize images to **640 × 640** (YOLOv8 default)
- Preserve aspect ratio using **letterboxing**
- Avoid aggressive blurring or compression
- Allow YOLOv8 to handle normalization internally

Why This Works

- Preserves sharp edges and characters
- Maintains spatial resolution for small objects
- Improves localization accuracy

---

5. Why did you choose YOLOv8 instead of a two-stage detector?

Comparison

Aspect	YOLOv8	Two-Stage Detectors
Architecture	Single-stage	Proposal + classification
Inference speed	Very fast	Slow
Real-time suitability	High	Limited
Deployment complexity	Low	High

Business Reasoning

This use case prioritizes:

- **Real-time processing**
- **Scalability**
- **Edge and production deployment**

YOLOv8 processes images in a **single forward pass**, making it ideal for high-throughput surveillance systems.

---

6. Which YOLOv8 variant did you select, and how did it balance speed and accuracy?

Selected Variant: YOLOv8-S

Variant	Speed	Accuracy	Verdict
YOLOv8-n	Very fast	Lower	Misses small plates
YOLOv8-s	**Balanced**	**High**	Selected
YOLOv8-m	Slower	Higher	Overkill
YOLOv8-l/x	Very slow	Marginal gain	Not cost-effective

Justification

YOLOv8-s provides sufficient model capacity to detect **small license plates** while maintaining real-time inference performance.

---

7. How did license plate characteristics influence your choice of model scale?

License plates are:

- Small relative to the image
- Rectangular and structured
- High-contrast but visually dense

These characteristics require:

- Strong spatial localization
- Multi-scale feature extraction
- Moderate model depth without overfitting

YOLOv8-s effectively balances these requirements without unnecessary computational cost.

---

8. How did you apply blurring only within the detected bounding box?

To preserve non-sensitive information, blurring is applied **exclusively inside the detected license plate region**.

Blurring Function

```
import cv2

def blur_license_plate(image, bbox):
    x1, y1, x2, y2 = bbox
    roi = image[y1:y2, x1:x2]
    blurred_roi = cv2.GaussianBlur(roi, (51, 51), 0)
    image[y1:y2, x1:x2] = blurred_roi
    return image
```

Business Importance

- Ensures complete anonymization of PII
- Preserves contextual information for analytics
- Meets privacy and regulatory requirements

Final Business Takeaway

This YOLOv8-based system delivers a **scalable, real-time, privacy-preserving solution** for license plate anonymization. By combining accurate coordinate transformations, high-quality annotations, efficient detection, and region-specific blurring, the solution achieves both **technical robustness** and **business compliance**.